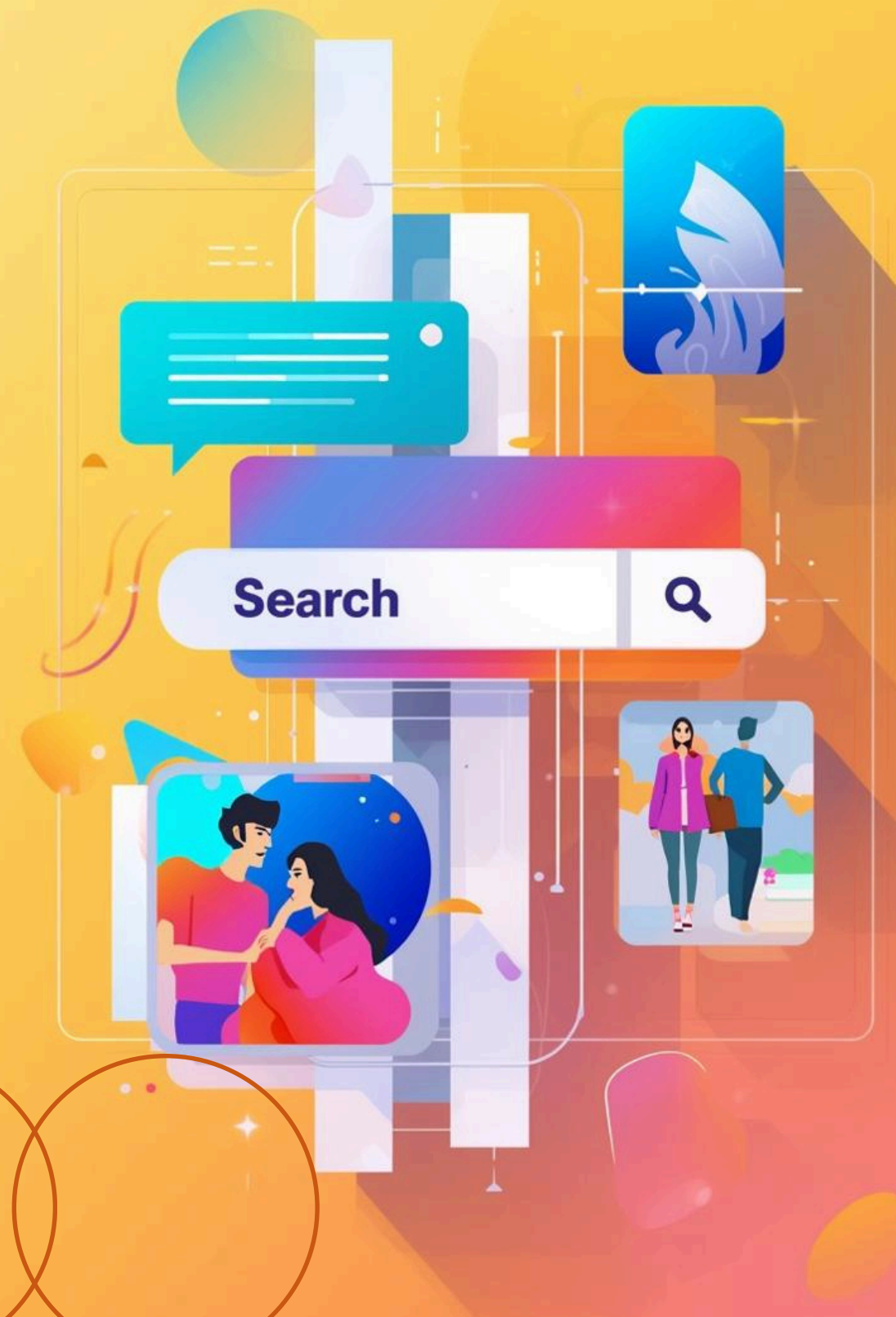
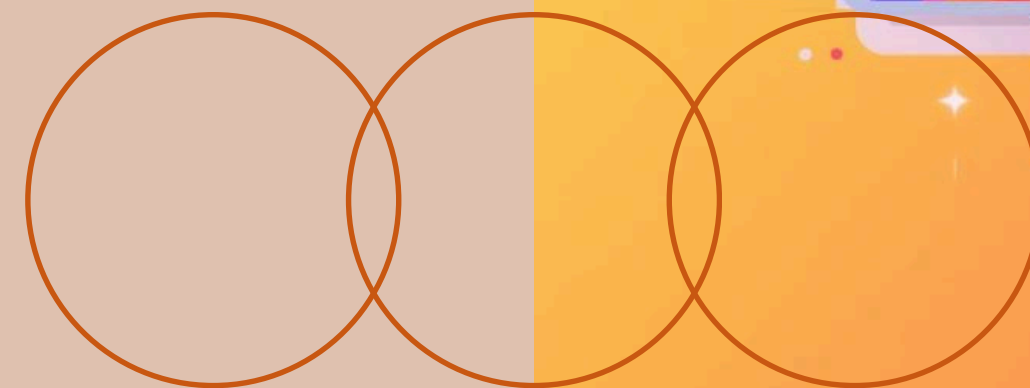


VECTORSEARCH

Multi-Modal Search Engine

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ITAI 1378 - COMPUTER VISION



Executive Summary

Bridging the "Vocabulary Gap" in E-Commerce Search

The Problem: Traditional keyword search fails when users can't describe visually complex items (e.g., specific patterns or styles).

The Solution: VectorSearch Pro, a multi-modal search engine that understands both text and images.

Technology Stack:

OpenAI CLIP: Converts images/text into a shared vector space.

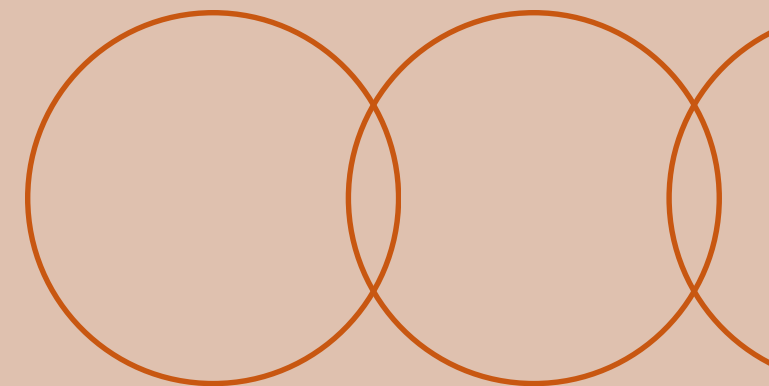
FAISS: Enables sub-millisecond similarity search.

Key Achievements:

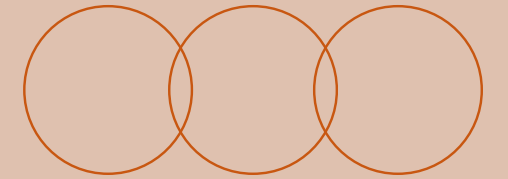
Scale: Indexed ~44,000 fashion items.

Performance: Achieved <0.3s latency (Goal: <1.0s).

Capabilities: Validated robust Text-to-Image and Image-to-Image retrieval.



Problem Statement



Addressing E-Commerce Challenges

Vocabulary Gap

The **vocabulary gap** arises when users input terms like "formal shirt" while metadata might list it as "button-down oxford," leading to irrelevant search results and user frustration.

Visual Feature Description

Complex visual features are often challenging to describe accurately with text alone, which limits user ability to find products that meet their aesthetic needs and expectations effectively.

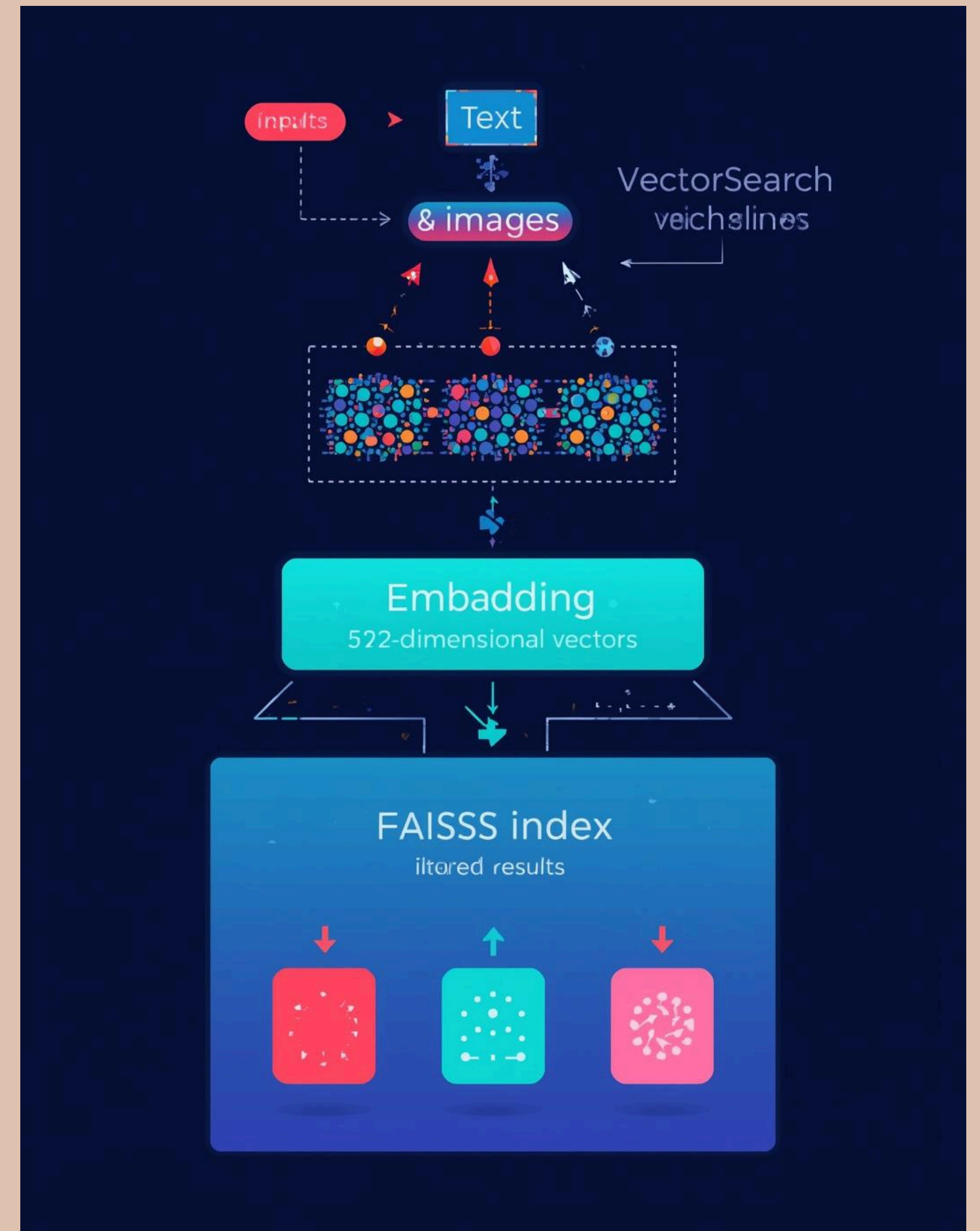
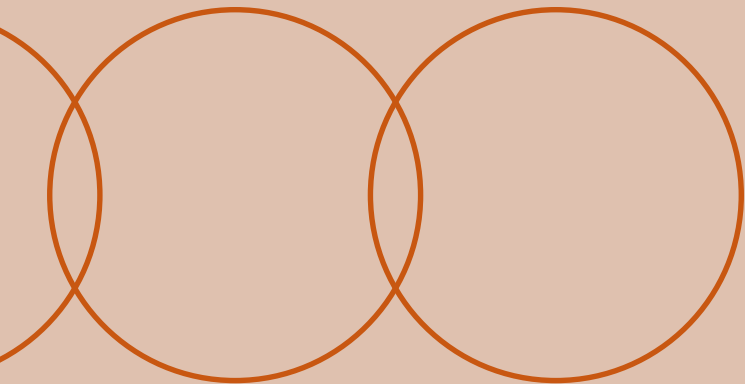
Image Queries

Supporting image queries allows users to leverage inspiration from social media photos, enhancing the search experience by tapping into visual context and enabling semantic understanding of products.

Architecture Overview

Understanding System Design

VectorSearch employs a **retrieve-and-rank pipeline** for effective search. It utilizes OpenAI's CLIP model for embedding, followed by FAISS for efficient indexing and retrieval.



Data Source and Preparation

Overview of Fashion Product Dataset

The dataset consists of over **44,400 fashion images**, meticulously linked to metadata for accurate categorization. This extensive collection enables enhanced search capabilities by providing diverse visual representations across numerous categories.

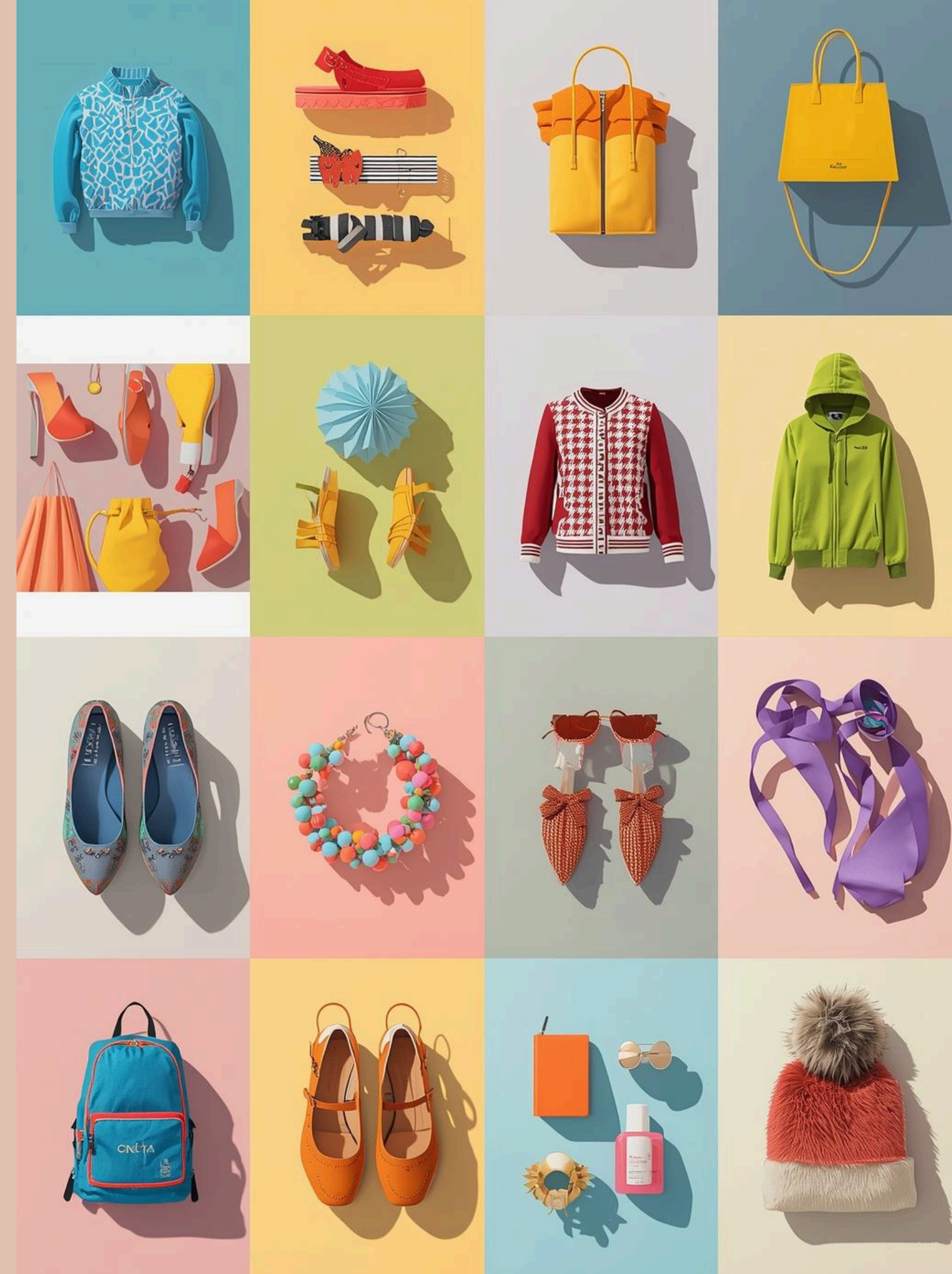
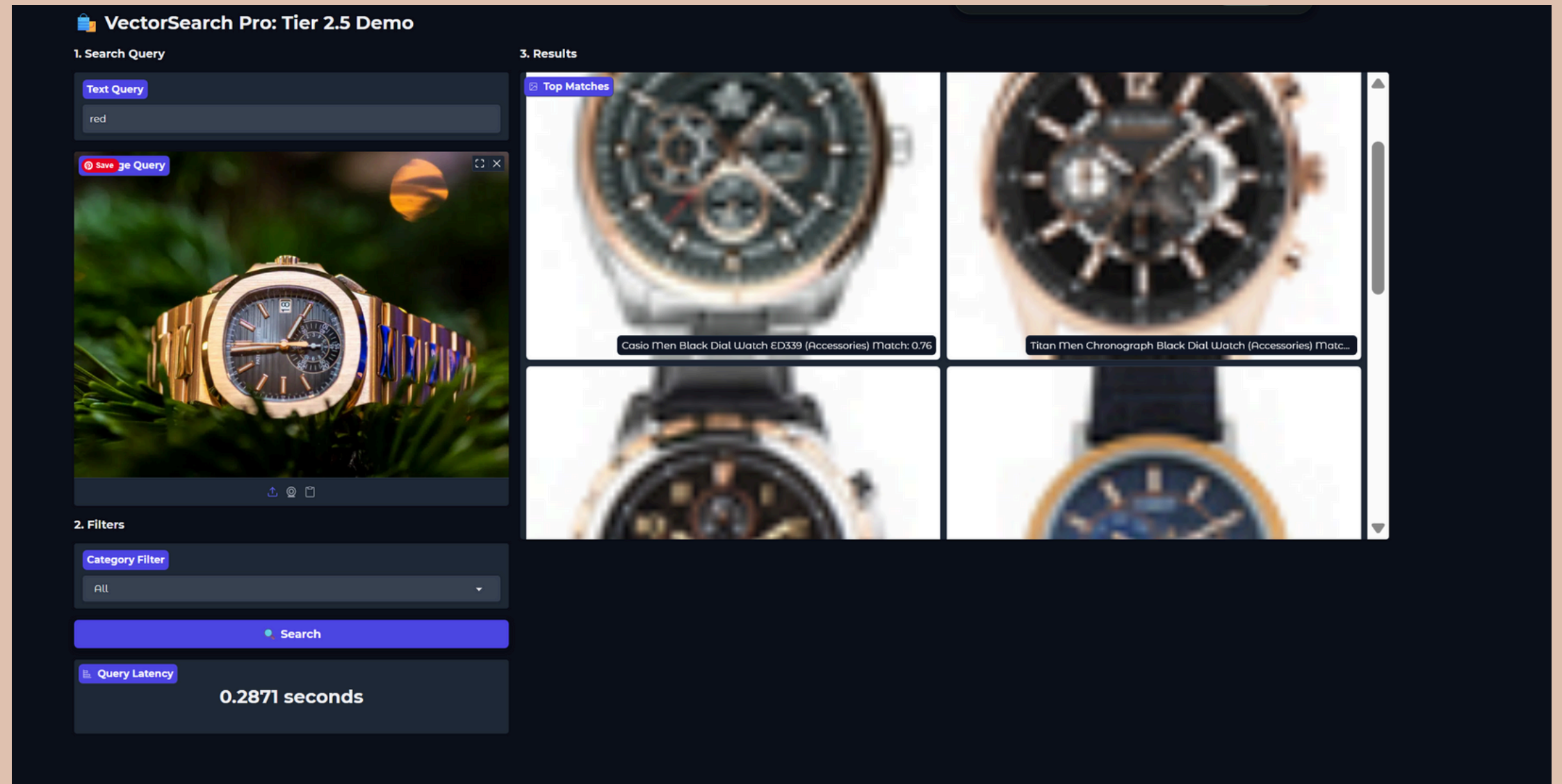


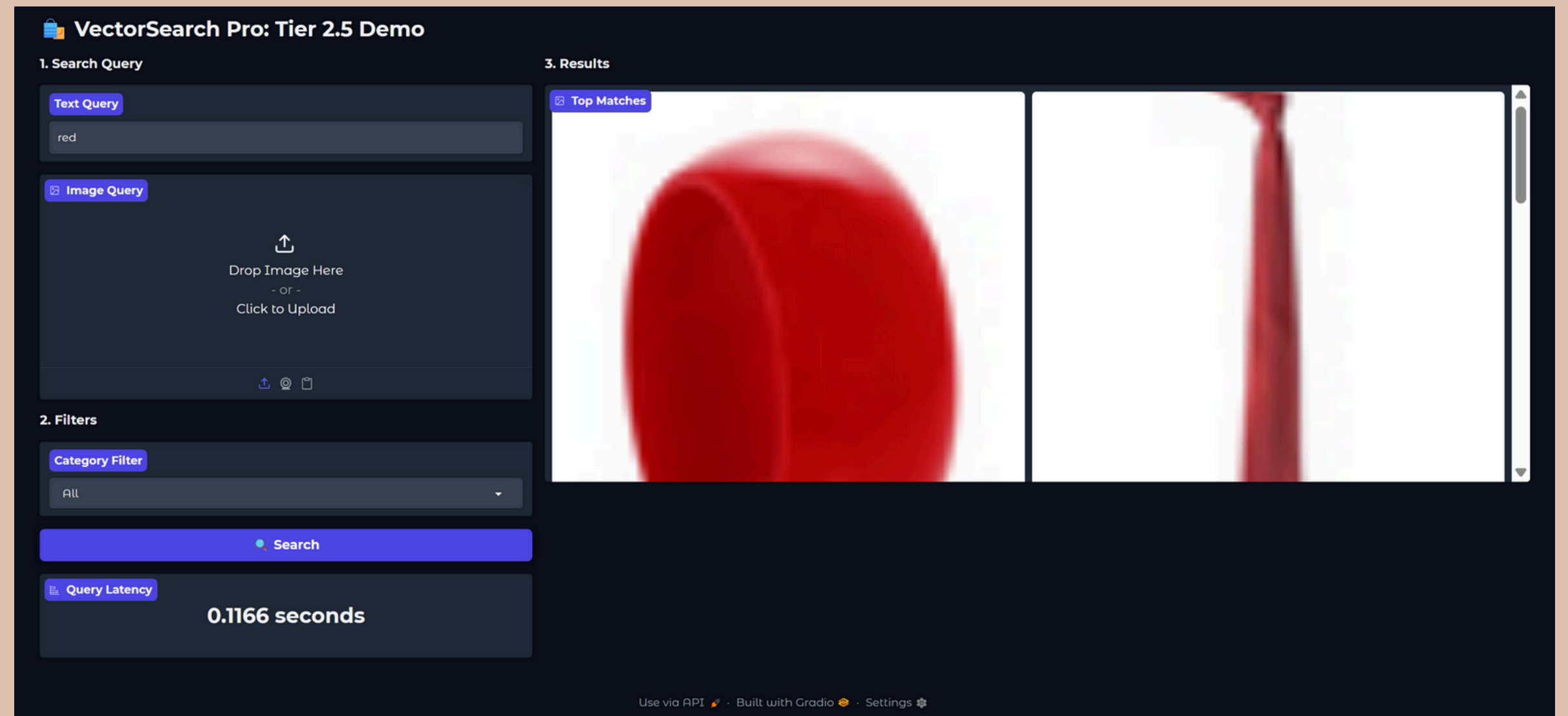
Image-to-Image Search



Luxury Watch Retrieval

The image search prioritized visual features (dial shape, metallic strap) over text, effectively functioning as a "find similar" tool.

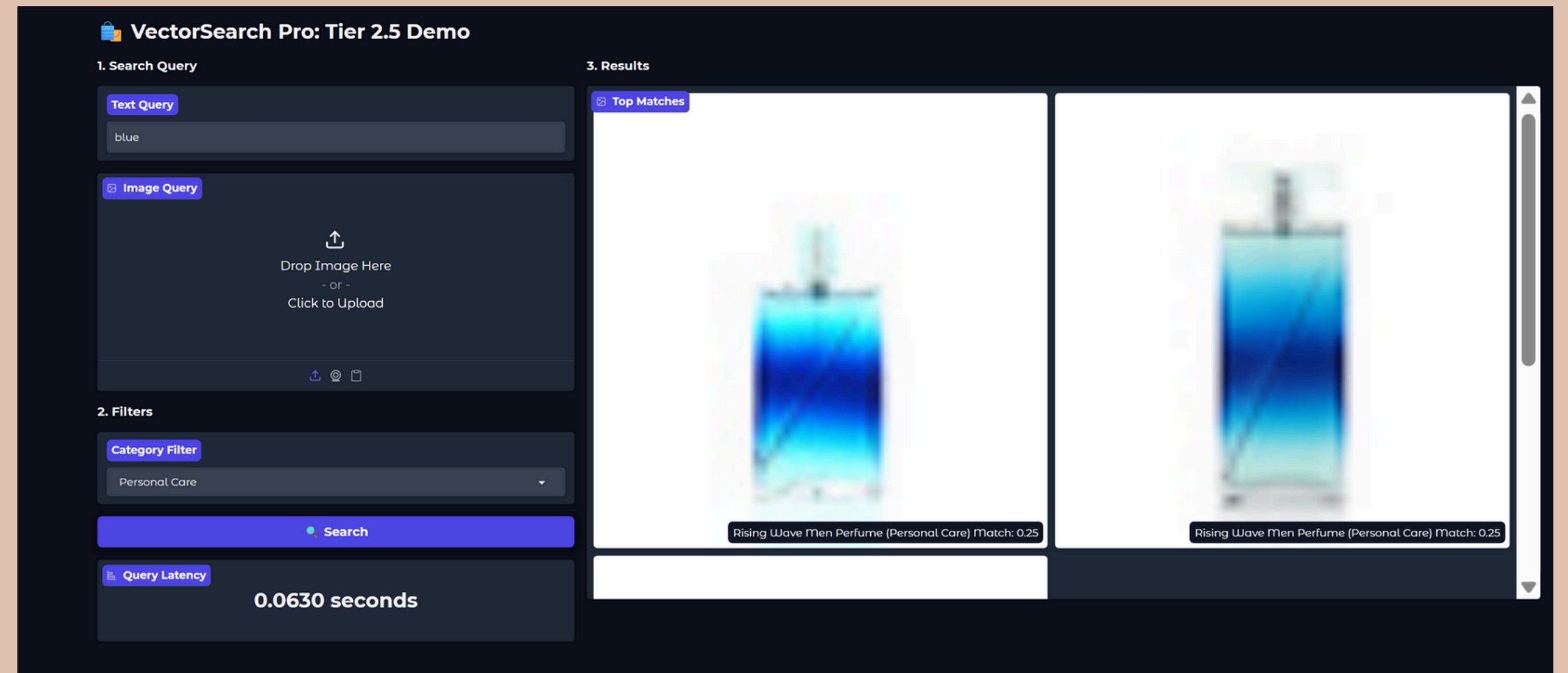
Text-to-Image Search: "Red"



Semantic Color Understanding

The model correctly linked the concept of "Red" to both jewelry and apparel, proving it understands color as an abstract property.

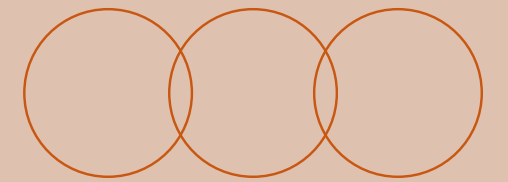
Text-to-Image Search: Scenario C



Filtered Blue Bottle Perfume

System correctly applied category filters to retrieve relevant 'Personal Care' items matching the color query.

Engineering Spotlight:



Solving the "Filter Gap" with Oversampling

The Challenge:

- Standard vector search finds the global top matches.
- If a user searches "Red" but filters for "Watches", and the global top 5 matches are all "Red Shirts", the result page appears empty.

The Solution:

- Implemented a custom Oversampling Strategy.
- Fetched 10x the requested candidates ($k * 10$) from the FAISS database.
- Applied category filtering locally on this larger set to ensure valid results are always returned to the user.

Code Snippet

```
# Fetch 10x candidates (k*10),  
then filter locally  
search_k = k * 10 if  
category_filter != "All" else k  
D, I = index.search(vec,  
search_k)
```

Performance Metrics

Validating system efficiency on a 44,000-image dataset.

- **Key Statistics (Big Numbers):**
 - 0.28s - Average Query Latency (vs. Target < 1.0s)
 - 44,000 - Total Images Indexed
 - Tier 2.5 - Project Complexity Classification
- **Latency Breakdown:**
 - Target Metric: 1.00s
 - Image Query: 0.28s
 - Text Query: 0.12s
- **Takeaway:** The system is approximately 3-8x faster than the initial performance target.

2. Filters

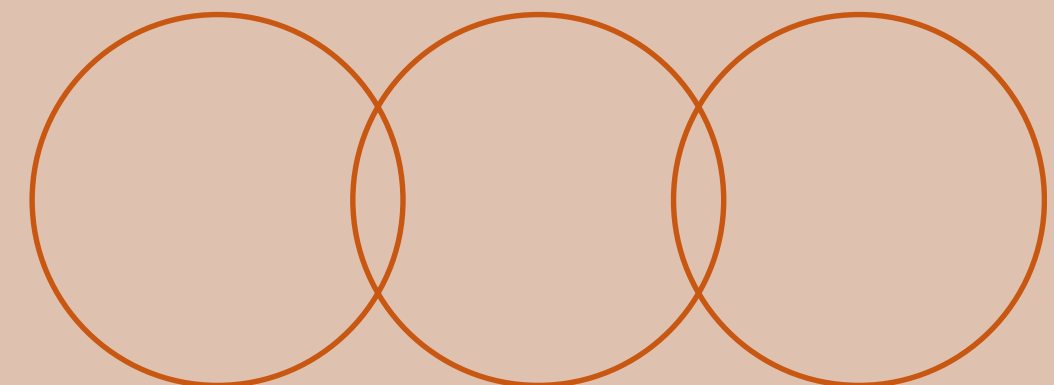
Category Filter

Personal Care ▼

 Search

Query Latency

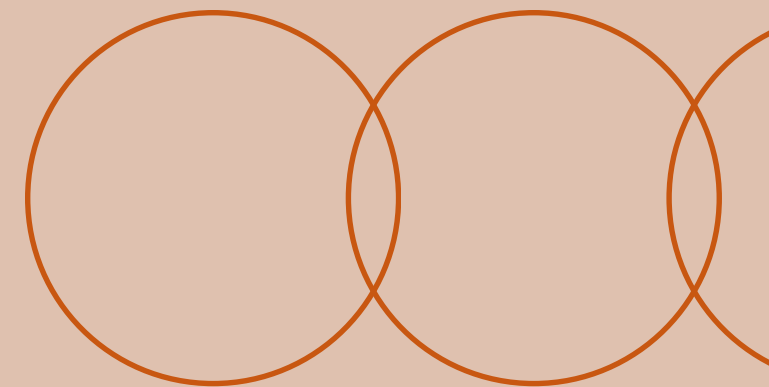
0.0360 seconds



Challenges

Challenges & Mitigations

- Challenge: Compute constraints on Colab GPUs.
- Mitigation: Implemented batch processing (size=32) and mixed precision (FP16) to optimize memory usage.
- Challenge: Data integrity issues (malformed CSV lines).
- Mitigation: Built robust error handling in the data pipeline to skip bad lines automatically.

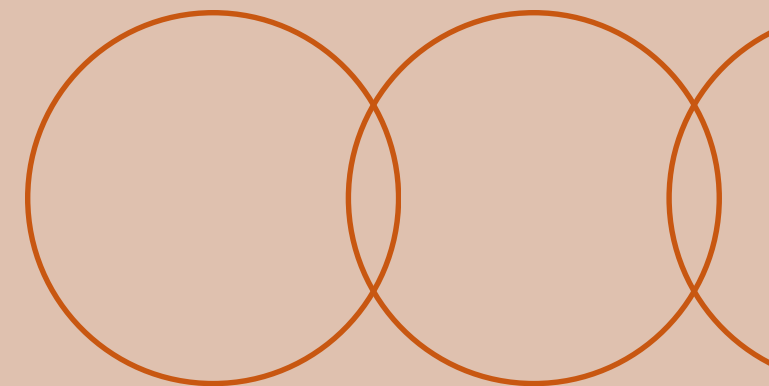


Future Enhancements for VectorSearch

Expanding capabilities and reach

Hybrid Search: Combine keyword matching with vector search for higher precision.

Fine-Tuning: Retrain CLIP on fashion-specific data to understand niche styles.



THANK YOU

Github

https://github.com/asamassekou10/ITAI-1378-FINAL_VectorSearch

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